

Yanyu Chen

Robotics (Mechanical Design/Embedded Systems) | Internship

Contact Information

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Location: Singapore

Summary

NUS Engineering undergraduate with dual-stack capabilities in mechanical design (CSWP certified) and embedded development. Independent full-stack robotic arm development experience and RoboMaster competition background. Able to communicate in English, Chinese, and German, adaptable to cross-national team collaboration.

Education Background

National University of Singapore (NUS)

Robotics and Machine Intelligence (B.Eng)

Minor: Innovation and Design | Class of 2029 | Singapore

Frankfurt International School

International Baccalaureate (IB Diploma)

2019.08-2025.05 | Germany | Score: 43/45 (Top 2% Global) | Honor: 2023-2025 highest distinction, Physics Award (Class of 2025)

Technical Skills

Mechanical Design | SolidWorks, Fusion 360, FEA, DFM/DFA

- 5+ year of CAD experience, engineering drawing

Embedded Systems | STM32 HAL/LL, Jetson Orin (CUDA), PID, Motor Control

- Experience in real-time control system development, familiar with CAN/RS485 communication

Robotics Related | ROS2, Gazebo, URDF, SLAM Navigation

Coding and Tools | Python, C++, CMake, Git, STM32CubeIDE, Keil

Experiences

NUS Calibur Robotics Competition Team (DJI RoboMaster Competition)

[Mechanical and Firmware team](#) | 2025.08 – present | Singapore

[Mechanical] “Engineer” robot 6dof arm design

- Adopted **truss topology** optimization and interlocking design, **reducing fasteners by 40%** and **reduced weight by 25%** compared to the previous generation
- Designed **internal cable routing** system and integrated conductive slip rings, enabling 360° continuous rotation of the end effector
- Applied DFA (Design for Assembly) principles through 4-5 mechanical iterations, combined 3D printing with CNC machining; the overall robotic arm features **quick-release structures** allowing any component to be fully replaced **within 3 min**

[Firmware] Embedded Platform-based Custom Controller

- Full-stack development of a custom controller (**STM32 platform**)
- Deployed Jetson Orin NX **edge computing platform**, configured and debugged sentry aimbot system

Ybot - Desktop 6-DOF Robotic Arm

Full stack Project | 2023.06 – 2025.08 | Personal Project | Mechanical + Firmware

- [Mechanical Design] Mechanical structural design in Fusion 360 (100+ CAD hours), achieved **30% weight reduction** through **FEA optimization**. Exported **URDF** for degree of movement simulation validation,
- [Hardware Architecture] Used **Raspberry Pi** as host computer, drove stepper motors via Arduino boards through CAN bus, completed distributed control system hardware setup

TurtleBot3 Simulated Warehouse Transportation Project (Mechanical / Firmware)

Coursework (CDE2310) | 2026.01 – 2026.04 | NUS

- Integrated **LiDAR** and vision to achieve SLAM mapping and path planning, completed static + dynamic station delivery tasks
- Developed **ArUco detection** and **automatic docking** system, achieving zero-angle alignment

Research Experience

Max-Plank Institute of Biophysics | Gerhard Hummer Lab

Student Researcher | 2024.06 – 2024.07 | Frankfurt – Germany

- Completed protein molecular dynamics simulations on HPC cluster using GROMACS engine, developed **Martini 3** coarse-grained modeling workflow
- Wrote automated analysis scripts in Python, achieving statistical processing and visualization of large-scale simulation data

Awards

2026 RoboMaster Highschool League (Chongqing) | Engineer Challenge | Second / First Price

HIMCM Mathematical Modeling Competition | M Award (Top 4 in Europe, 2023)

Certification

- SolidWorks CSWP (Full Score)

Language

English: **Fluent** (Academic/ Work)

Chinese: **Mother tongue**

German: **Proficient** (TELC C1 certificate)